

Homework 2

EE 999 — Introduction to Control Systems

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Due Wednesday, September 23, 2026

Problem 1.A.2.1 — What is a "free-body diagram"?

A free body diagram is a diagram that shows the forces acting on an object or set of objects.

Problem 1.A.2.2 — What are the two forms of Newton's law?

Translational motion: $F = ma$; Rotational motion: $M = I\alpha$

Problem 1.A.2.3 — For a structural process to be controlled, such as a robot arm, what is the meaning of "collocated control" and "Noncollocated control"?

Collocated control means the sensor and actuator are rigidly attached to one another, whereas there is flexibility between the actuator and sensor in a noncollocated system.

Problem 1.A.2.4 — State Kirchhoff's current law.

Kirchhoff's current law states that current entering any given electrical node must equal current exiting a node.

Problem 1.A.2.5 — State Kirchhoff's voltage law.

Kirchhoff's voltage law states that the sum of all voltages around a closed circuit must equal zero.

Problem 1.B — Differential equation for mechanical system

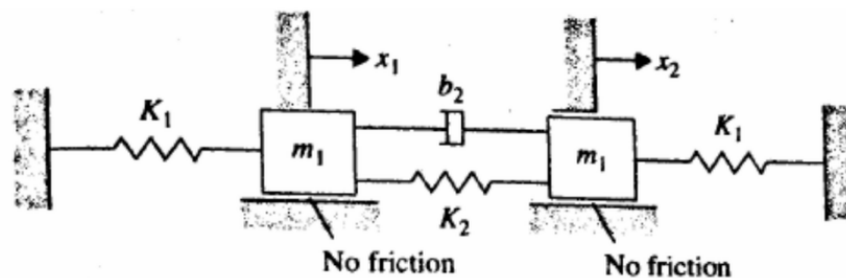


Figure 1: The two-mass system of Problem 1.B.

$$A = b + a$$

The difference in velocities between m_1 and m_2 will decay because b_2 scales with velocity. Since there is no friction, the output $x_2(t)$ will not decay.